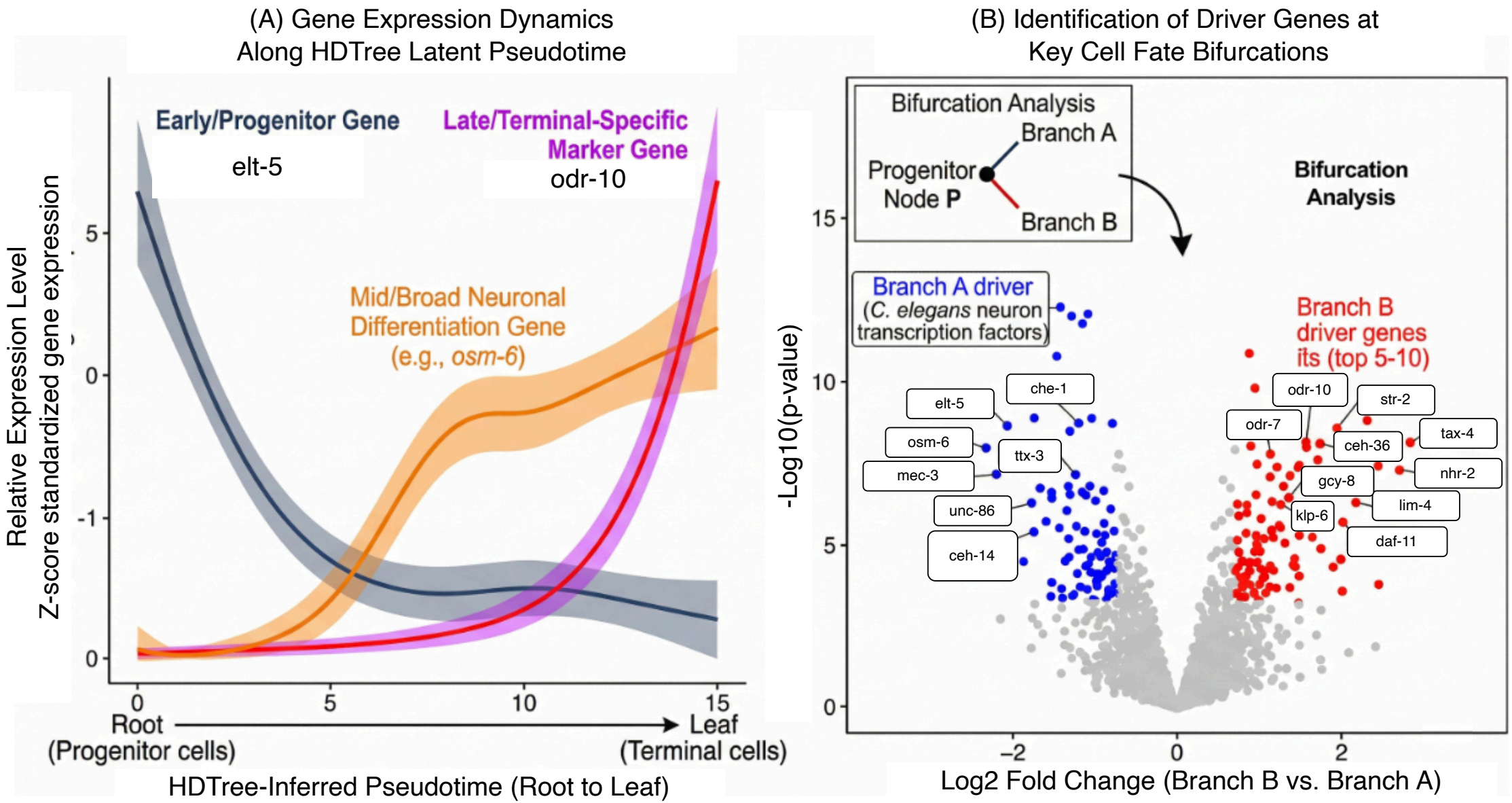

Figure R3: Biological interpretability and lineage validation of HDTree on the C. elegans dataset. (A) Gene Expression Dynamics: Expression profiles along the HDTree-inferred pseudotime. The model accurately captures the temporal progression of early progenitor (elt-5), intermediate (osm-6), and late terminal (odr-10) marker genes. (B) Bifurcation Analysis: Differential expression (volcano plot) at a key cell fate bifurcation. HDTree's explicit branching structure successfully separates and identifies distinct driver genes for diverging sibling lineages (Branch A vs. Branch B).